



MAR <maramalraddadi17@gmail.com>

Classif_PretrainedCNN

1 message

Muslim Alhammad <muslimalhammad@gmail.com>

Sun, Sep 6, 2026 at 6:25 PM

To: Muslim Alhammad <muslimalhammad@gmail.com>, Haya Saleh <Hayascareer@gmail.com>, maramalraddadi17@gmail.com, raed7601@gmail.com

Downloading data from https://storage.googleapis.com/keras-applications/efficientnetb0_notop.h5
16705208/16705208 ————— 2s 0us/step

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0
efficientnetb0 (Functional)	(None, 7, 7, 1280)	4,049,571
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 256)	327,936
dropout_1 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 4)	1,028

Total params: 4,378,535 (16.70 MB)**Trainable params:** 328,964 (1.25 MB)**Non-trainable params:** 4,049,571 (15.45 MB)

Epoch 1/6

250/250 ————— 0s 20ms/step - accuracy: 0.8774 - loss: 0.3233

Epoch 1: val_accuracy improved from None to 0.91100, saving model to /tmp/best_model.keras

Epoch 1: finished saving model to /tmp/best_model.keras

250/250 ————— 262s 1s/step - accuracy: 0.8802 - loss: 0.3172 - val_accuracy: 0.9110 - val_loss: 0.2320

Epoch 2/6

248/250 ————— 0s 19ms/step - accuracy: 0.8973 - loss: 0.2703

Epoch 2: val_accuracy improved from 0.91100 to 0.92500, saving model to /tmp/best_model.keras

Epoch 2: finished saving model to /tmp/best_model.keras

250/250 ————— 67s 30ms/step - accuracy: 0.8915 - loss: 0.2823 - val_accuracy: 0.9250 - val_loss: 0.1994

Epoch 3/6

250/250 _____ **0s** 20ms/step - accuracy: 0.9133 - loss: 0.2247

Epoch 3: val_accuracy did not improve from 0.92500

250/250 _____ **6s** 25ms/step - accuracy: 0.9022 - loss: 0.2403 - val_accuracy: 0.9220 - val_loss: 0.1971

Epoch 4/6

250/250 _____ **0s** 20ms/step - accuracy: 0.9199 - loss: 0.2126

Epoch 4: val_accuracy did not improve from 0.92500

250/250 _____ **6s** 25ms/step - accuracy: 0.9162 - loss: 0.2169 - val_accuracy: 0.9140 - val_loss: 0.2119

Epoch 5/6

250/250 _____ **0s** 20ms/step - accuracy: 0.9160 - loss: 0.1967

Epoch 5: val_accuracy did not improve from 0.92500

250/250 _____ **6s** 26ms/step - accuracy: 0.9160 - loss: 0.2003 - val_accuracy: 0.9240 - val_loss: 0.1848

Epoch 6/6

250/250 _____ **0s** 19ms/step - accuracy: 0.9318 - loss: 0.1831

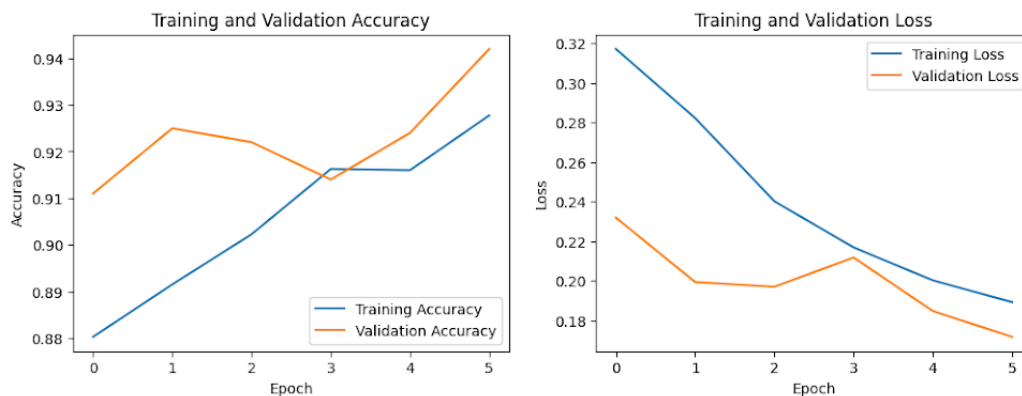
Epoch 6: val_accuracy improved from 0.92500 to 0.94200, saving model to

/tmp/best_model.keras

Epoch 6: finished saving model to /tmp/best_model.keras

250/250 _____ **7s** 27ms/step - accuracy: 0.9277 - loss: 0.1893 - val_accuracy: 0.9420 - val_loss: 0.1718

Model training complete. Best model saved to: /tmp/best_model.keras



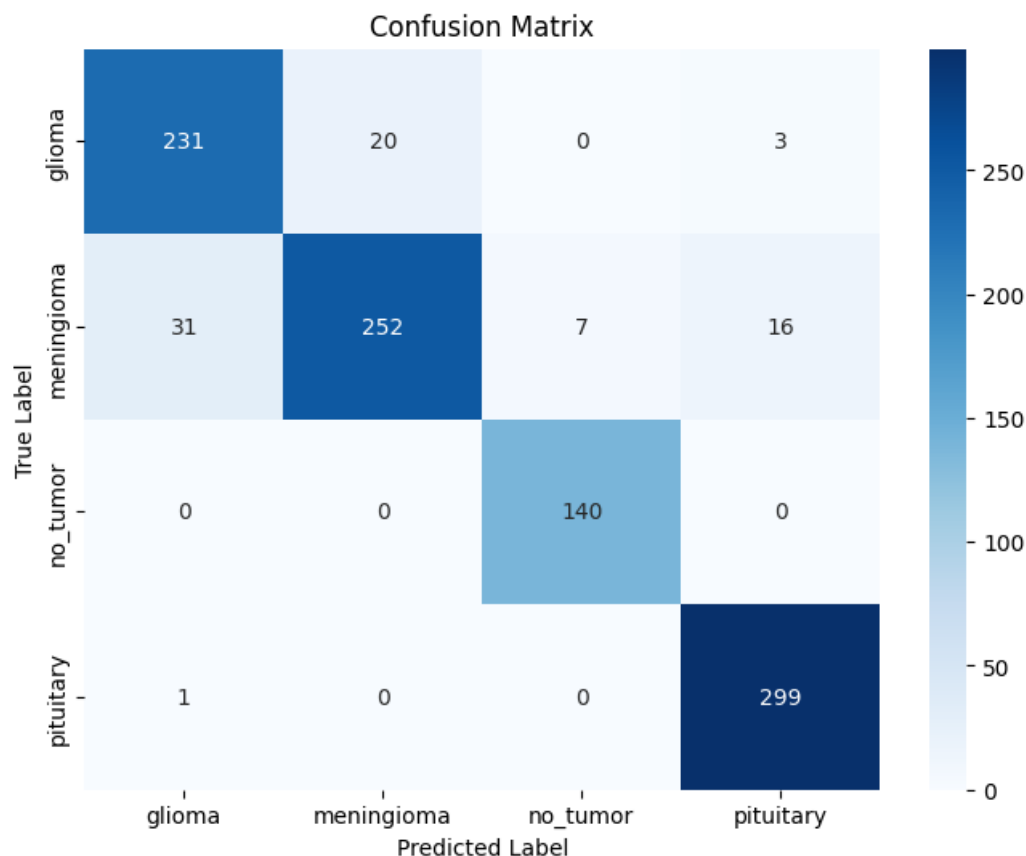
Evaluating model on the test dataset...

63/63 _____ **298s** 5s/step - accuracy: 0.9220 - loss: 0.1897

Test Loss: 0.1897

Test Accuracy: 0.9220

63/63 _____ **15s** 145ms/step

**Classification Report:**

	precision	recall	f1-score	support
glioma	0.88	0.91	0.89	254
meningioma	0.93	0.82	0.87	306
no_tumor	0.95	1.00	0.98	140
pituitary	0.94	1.00	0.97	300
accuracy			0.92	1000
macro avg	0.92	0.93	0.93	1000
weighted avg	0.92	0.92	0.92	1000

1/1 0s 54ms/step

